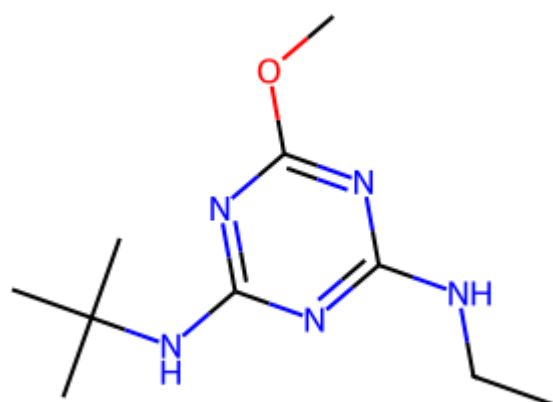
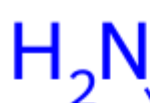




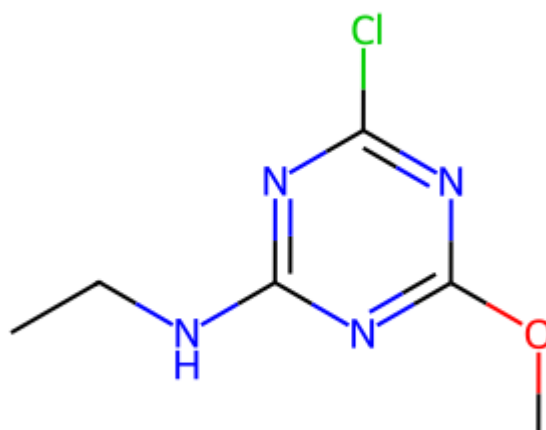
{'analysis_time': 0.0,
'first_solution_time': 0.607988,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 3.2051075441452888,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 3.2051075441452888,
'retro_star_rxn_cost': 2.441847160327553,
'retro_star_value': 3.2051075441452888}



Reaction
{'analysis_time': 0.172091,
'first_solution_time': 0.610266,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 5.107527765063254,
'retro_star_reaction_number': 5.107527765063254,
'retro_star_rxn_cost': 2.441847160327553,
'retro_star_value': 5.107527765063254}



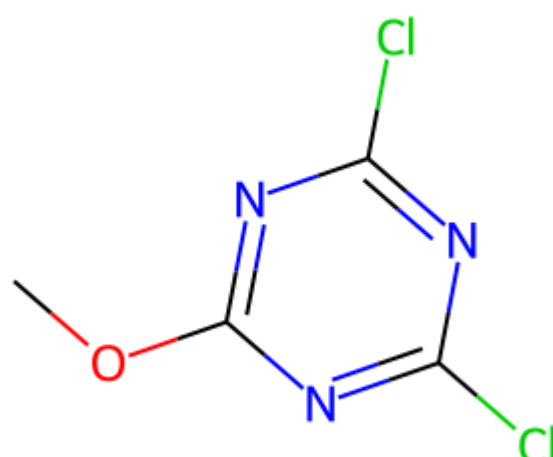
{'analysis_time': 0.172091,
'first_solution_time': 0.172091,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 5.107527765063254}



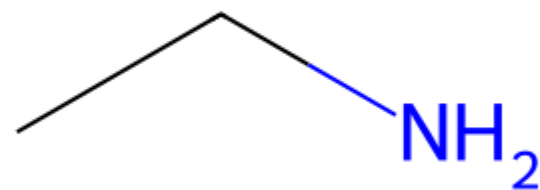
{'analysis_time': 0.172091,
'first_solution_time': 0.610266,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 2.6656806047357007,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 2.6656806047357007,
'retro_star_rxn_cost': 0.4400565528777834,
'retro_star_value': 5.107527765063254}



Reaction
{'analysis_time': 0.478265,
'first_solution_time': 0.610266,
'num_calls_rxn_model': 4,
'num_calls_value_function': 92,
'retro_star_min_cost': 2.6656806047357007,
'retro_star_reaction_number': 2.6656806047357007,
'retro_star_rxn_cost': 0.4400565528777834,
'retro_star_value': 5.107527765063254}



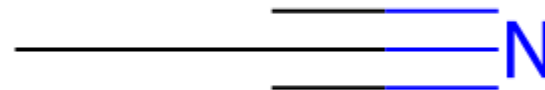
{'analysis_time': 0.478265,
'first_solution_time': 0.478265,
'num_calls_rxn_model': 4,
'num_calls_value_function': 92,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 5.107527765063254}



{'analysis_time': 0.478265,
'first_solution_time': 0.610266,
'num_calls_rxn_model': 4,
'num_calls_value_function': 92,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 2.2256240518579173,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 2.2256240518579173,
'retro_star_value': 5.107527765063254}



Reaction
{'analysis_time': 0.610266,
'first_solution_time': 0.610266,
'num_calls_rxn_model': 5,
'num_calls_value_function': 174,
'retro_star_min_cost': 2.2256240518579173,
'retro_star_reaction_number': 2.2256240518579173,
'retro_star_rxn_cost': 2.2256240518579173,
'retro_star_value': 5.107527765063254}



{'analysis_time': 0.610266,
'first_solution_time': 0.610266,
'num_calls_rxn_model': 5,
'num_calls_value_function': 174,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 5.107527765063254}